

Day 29 — Random Forest Classifier

Veda Technology · Level 2 · Day 29 · Data Science Track

Description: Build a Random Forest classification model and compare it with a single Decision Tree.

Objective: Understand ensemble learning and why combining multiple trees can improve generalization.

Dataset: Breast Cancer Wisconsin Dataset from scikit-learn.

1. Workflow

- Load and inspect data
- Train-test split
- Decision Tree baseline
- Random Forest model
- Compare performance
- Experiment with `n_estimators`
- Feature importance
- Classification report and confusion matrix
- Generalization analysis

2. Random Forest Code

```
rf = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42,  
    n_jobs=-1  
)  
rf.fit(X_train, y_train)
```

3. `n_estimators` Experiment

The notebook tests 10, 25, 50, 100, 200 and 300 trees and compares training and testing accuracy.

4. Feature Importance

The Random Forest `feature_importances_` attribute is used to rank the most important features.

5. Overfitting and Generalization

Training accuracy is compared with testing accuracy. A large gap can indicate overfitting, while test performance indicates how well the model generalizes.

6. Interview Questions

What is a Random Forest? An ensemble of many decision trees.

Why does it usually generalize better? Combining trees generally reduces variance.

What is ensemble learning? Combining multiple models to create a stronger prediction.

What does `n_estimators` control? The number of trees.

7. Conclusion

Day 29 demonstrates Random Forest classification, comparison with a Decision Tree, estimator tuning, feature importance, and generalization analysis.